

Detection, Segmentation and Calibration of Rubber Strips

Computer Vision

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1. Introduction

What is the problem?

- Detection, segmentation and distance estimation of rubber strips from *smartphone images*

What is it challenging?

- Real-world smartphone images
- Precise distance estimation
- Perspective, noise, variability

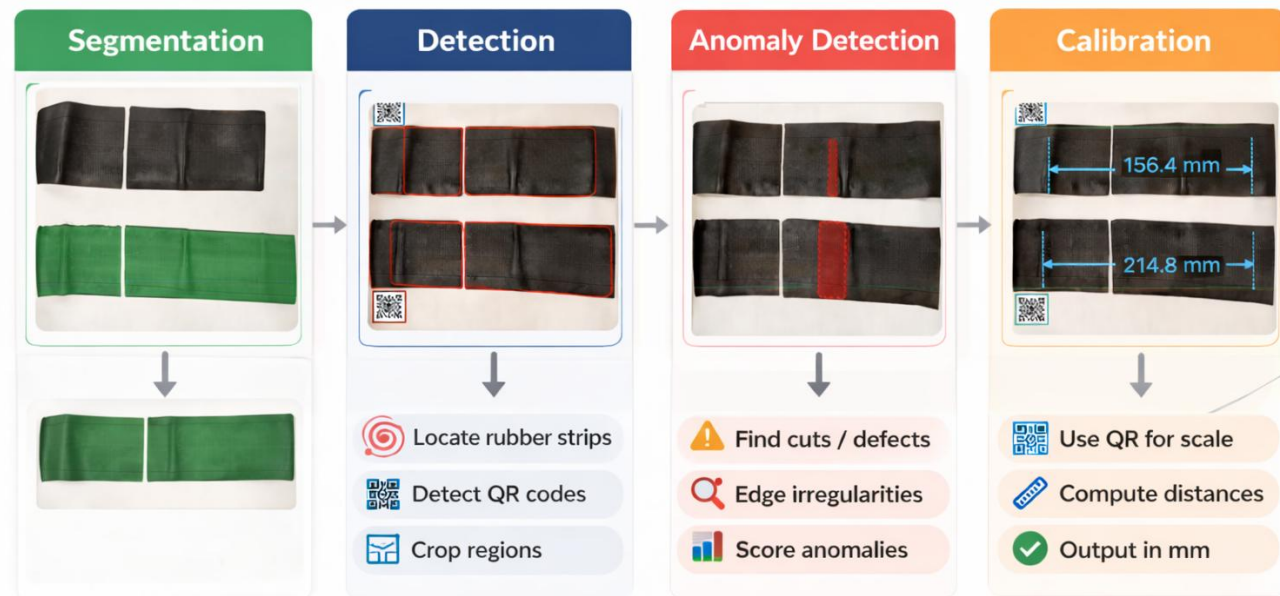


Solution: *Classical vs Deep Learning*

1. Introduction: *State of the art*

Main approaches

- *Segmentation*
- *Detection*
- *Anomaly Detection*
- *Calibration*



2. Literature review – Deep Learning

Main method categories: **Segmentation**, **Detection** and **Anomaly Detection**

Method	Year	Dataset	Main Metric	Code	Speed
YOLOv8	2024	COCO, Roboflow 100	71.5% mAP@0.5	Yes	11.5 ms
RT-DETRv4	2025	COCO 2017	49.7–57.0 AP	Yes	78–273 FPS
Segformer++	2024	Cityscapes, ADE20K	82.31 mIoU	Yes	1.61× speedup
PEM	2024	Cityscapes, ADE20K	61.1 PQ / 79.9 mIoU	Yes	13.8–35.7 FPS
Relation-DETR	2024	COCO 2017	51.7 AP	Yes	N/R
ViLU-Net	2025	CT Tumour, FLARE	DSC 0.9309 / IoU 0.8720	Yes	N/R
Grounded-SAM	2024	SegInW (zero-shot)	48.7 mAP	Yes	N/R
YOLO-World	2024	LVIS	35.4 AP	Yes	52.0 FPS
Grounding DINO 1.5	2024	COCO, LVIS	54.3 AP / 55.7 AP	Yes	5.5–75.2 FPS
AnoDDPM	2024	LUVT	AUROC 0.909	Yes	N/R

2. Literature review – Deep Learning

Main method categories: **Segmentation**, **Detection** and **Anomaly Detection**

Method	Year	Dataset	Main Metric	Code	Speed
SAM 2	2024	MOSE val, SA-23	J&F 76.6% / mIoU 61.9%	Yes	43.8 FPS
EfficientSAM	2023	COCO, LVIS	44.4 AP (zero-shot)	Yes	47–54 img/s
YOLOv9	2024	MS COCO	55.6% AP_{50:95}	Yes	N/R
YOLOv8-HDSA	2025	Industrial carbon block	+7.2% accuracy over YOLOv8	Yes	N/R
AnomalyDINO	2024	MVTec-AD	AUROC 96.6% (1-shot)	Yes	N/R
CLIP-DINOv2	2025	MVTec AD, VisA	AUROC 93.4% / AP 94.3%	No	23 FPS
SAM2-UNet	2026	18 datasets (5 benchmarks)	mDice 92.8% (Kvasir)	Yes	N/R
Semi-sup. defect seg.	2024	KolektorSDD, Steel	mIoU +4.39% over SOTA	No	64.59 FPS
Camera Calib. Survey	2025	Holistic calib. dataset	N/R (survey)	Yes	N/R
MVTec AD 2	2025	MVTec AD 2	AU-PRO <60% (SOTA)	No	N/R

2. Literature review – Classical

Main method categories: **Segmentation**, **Detection** and **Anomaly Detection**

Method	Year	Dataset	Main Metric	Code	Speed
Modified Watershed	2025	LiTS17	PSNR, SNR, MSE (improved)	Yes	N/R
MAMDD	2025	BSDS500, NYUDv2, PASCAL VOC	PR curves, FOM	No	N/R
Fuzzy Segmentation	2025	Segmentation datasets	96.8% accuracy, F1: 95.1%	Yes	18.5 s
SPORG-RBVS	2024	DRIVE, STARE, CHASE_DB1	98.68% accuracy	No	N/R
Ridge Segmentation	2025	ABC-10, OmniObject3D-8	95.52 mIoU	No	N/R
HOG + LBP + Entropy + SVM	2025	CIFAR-10, Fashion-MNIST	94.81% accuracy	Yes	N/R
Harris + Bezier + Delaunay	2025	3D object modeling	120 pts / 3.2 s	Yes	3.2 s
Subpixel Edge Detection	2024	Industrial measurement	Reduced absolute/relative error	No	N/R
Industrial Pipeline	2025	Industrial datasets	N/R	No	N/R
Edge Detection Review	2024	Literature	Precision, Recall, F1	No	N/R

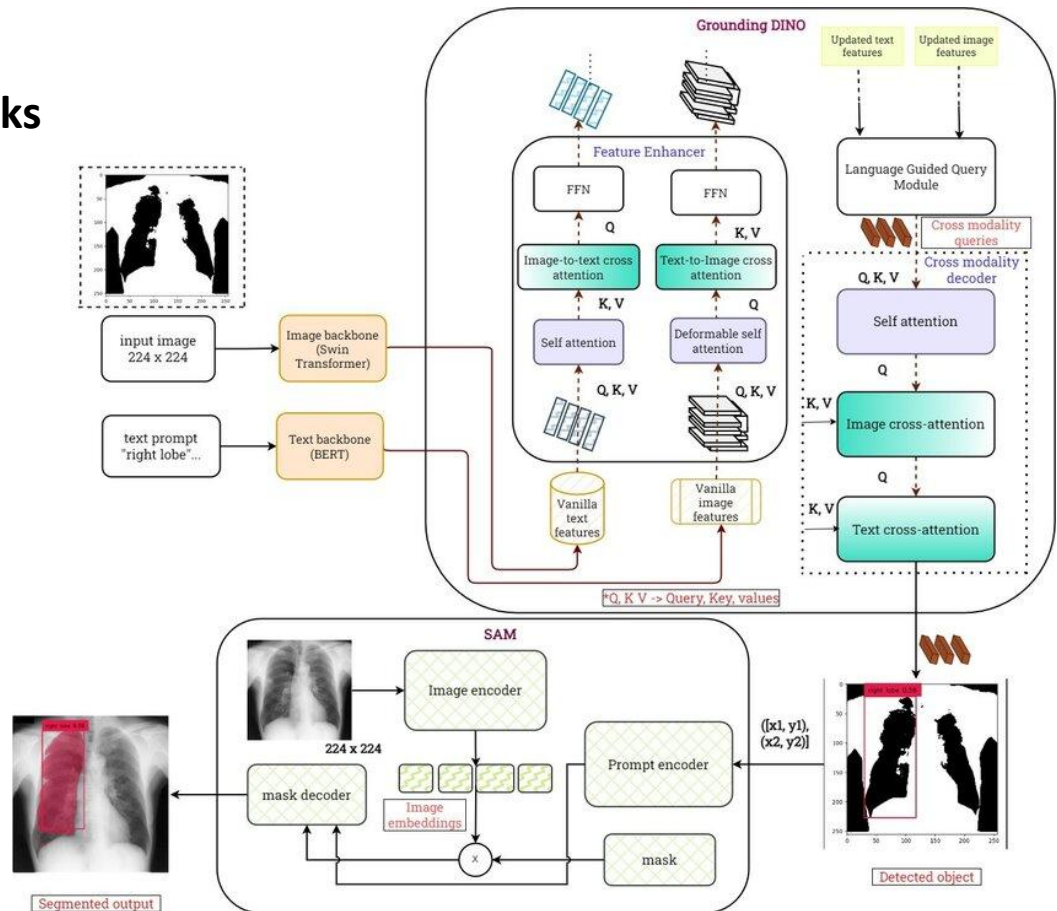
2. Literature review – Classical

Main method categories: **Segmentation**, **Detection** and **Calibracion**

Method	Year	Dataset	Main Metric	Code	Speed
HOG + SVM / PCA + RF	2024	Olive fly dataset	Accuracy: 99%, F1: 0.96	No	9 ms/ROI
Adaptive Edge Finishing	2024	Aeronautical parts	RMSE: 0.383 mm	No	31.7 s/image
HOG / LBP / SIFT + KNN	2025	10,580 X-ray images	Accuracy: 99.03%	No	N/R
Grid-HOG-PCA-LightGBM	2024	Industrial surface (399 images)	Accuracy: 98%	Yes	0.05 s
HOG + SVM / XGBoost	2025	40,000 crack images	Accuracy: 99.05%	No	N/R
GLCM + SVM / CNN	2024	Tool wear dataset	Accuracy: 93%+ (CNN)	No	N/R
HOG + PCA + Ensemble	2024	PRPD dataset	Accuracy: 89.79%	No	N/R
Improved Canny	2024	Rebar dataset	AP: 0.92	Yes	N/R
Homography Decomp.	2025	Benchmark dataset	4–7× faster than SVD	Yes	4–7×
Camera Calibration	2024	Checkerboard dataset	Improved accuracy	Yes	N/R
Canny + Hu + SVM	2025	Automotive flange	Accuracy: 97.2%	No	N/R

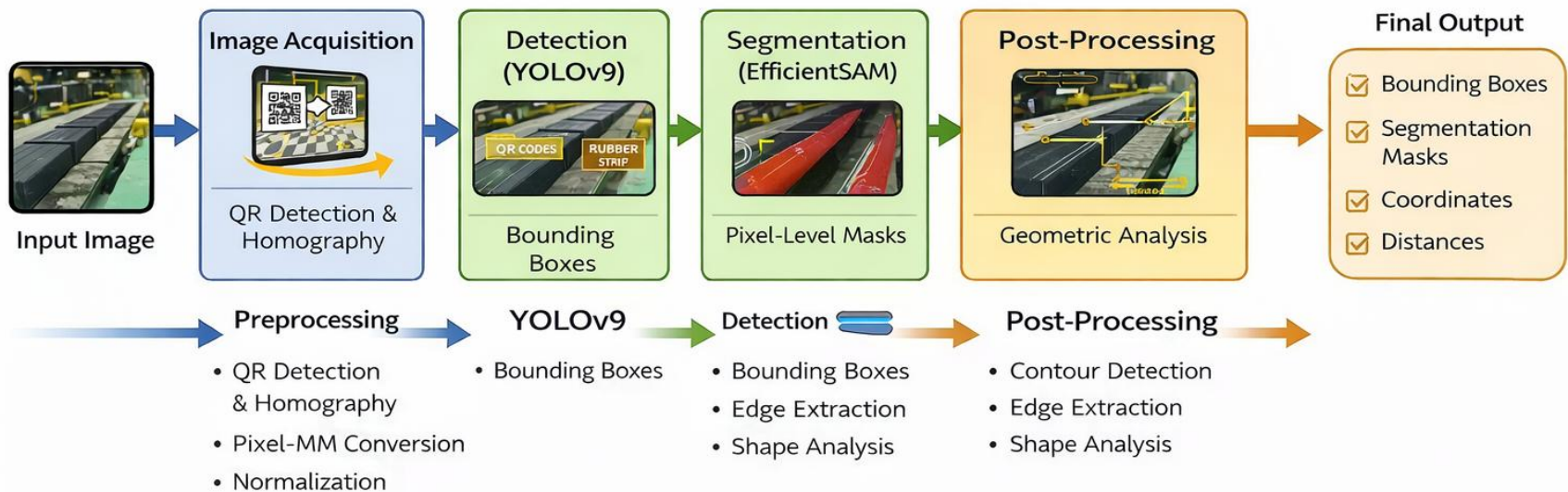
3. Selected method (Deep 1): Grounded-SAM

- Detection + segmentation pipeline (2-stage)
- Open-vocabulary
- Object detection
- Precise segmentation masks
- Zero-shot



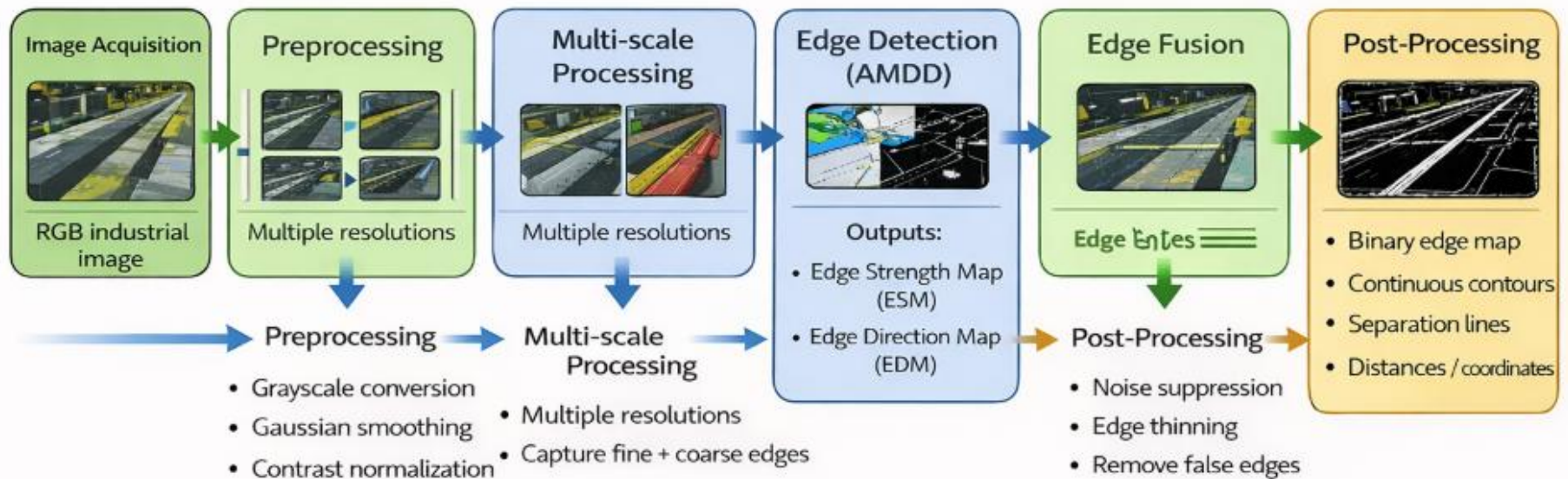
3. Selected method (Deep 2): YOLOv9 + EfficientSAM

- Detection + segmentation pipeline (2-stage)
- Real-time object detection (YOLOv9)
- Lightweight segmentation (EfficientSAM)
- QR-based calibration + homography
- Precise Edge localization and measurement
- Robust to real-world conditions



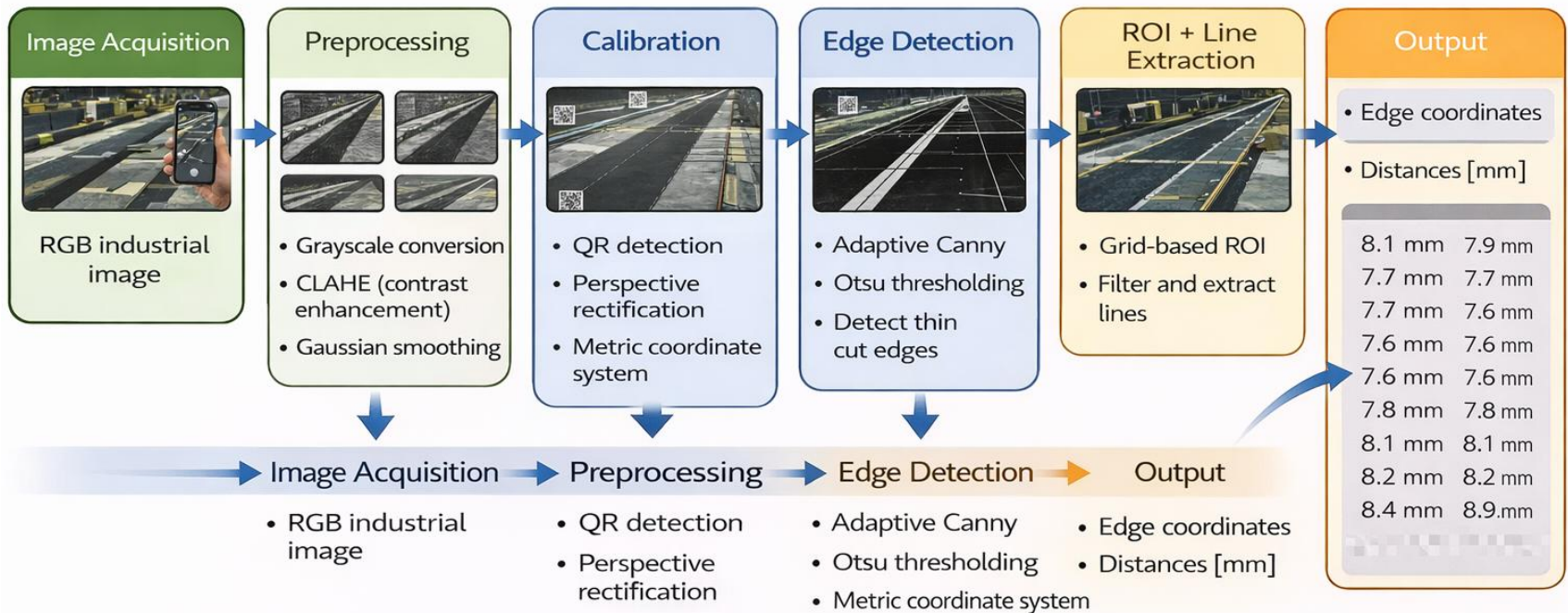
3. Selected method (Classical 1): AAMD

- Multi-scale edge detection pipeline (classical)
- Noise-robust edge extraction (AMDD)
- Anisotropic morphological filtering
- Directional edge analysis (ESM + EDM)
- Edge fusion for continuity and accuracy
- Robust to low contrast and industrial noise



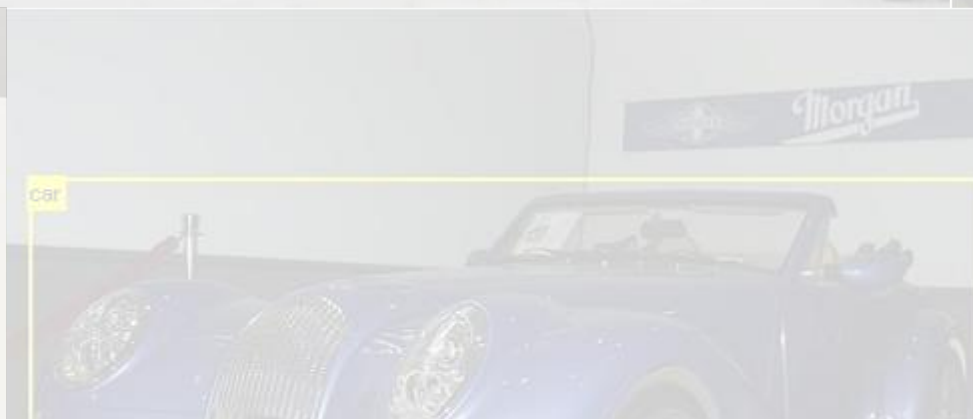
3. Selected method (Classical 2): Adaptative Canny

- Adaptive Canny edge detection (Otsu-based)
- QR-based homography calibration (mm estimation)
- Perspective correction + lens distortion removal
- ROI filtering + Hough line extraction
- Geometric filtering and line merging
- Robust to illumination, perspective and noise
- No training required (CPU-based deployment)



4. Datasets

Dataset	Year	Task	Key Info
COCO	2014	Detection / Segmentation	330K images, 80 everyday object categories
NEU Defect	2013	Defect detection	1800 steel surface images, 6 defect types
LVIS	2019	Detection / Segmentation	1203 categories, long-tail distribution
SA-1B	2023	Segmentation	11M images, 1.1B masks (large-scale)
BSDS500	2011	Edge detection / Segmentation	500 images with human edge annotations
PASCAL VOC 2007	2007	Detection	9963 images, 20 object classes
IP102	2019	Classification / Detection	75K images, 102 insect pest categories
MURA	2017	Classification	40K X-ray images, normal vs abnormal
Michelin Dataset	2026	Industrial measurement	QR-based calibration + rubber strips



5. Performance Metrics

◆ Detection & Segmentation

- Precision / Recall / F1-Score
- IoU (overlap quality)
- AP / mAP (overall detection performance)

◆ Measurement

- RMSE / MAE
- Distance Error

◆ Edge Detection

- Precision-Recall Curve
- FOM (Edge accuracy)

6. Conclusions

- **Deep learning** highest accuracy, but requires **data + GPU**.
- **Classical methods** efficient and no data needed, but less robust to real conditions.
- **Grounded-SAM** is the most flexible approach, enabling **zero-shot detection** and **segmentation**, making it suitable when labeled data is unavailable.
- **YOLOv9 + EfficientSAM** provides the best **accuracy-speed trade-off**, but depends on **fine-tuning** and **training data**.
- **MAMDD** enhances classical edge detection with **robustness to noise and low contrast**, although it still relies on handcrafted processing.
- **Adaptive Canny pipeline** is the most **practical and deployable solution**, requiring **no training and only CPU**, but with lower robustness.

**Thank you for your
attention**

7. References

- [1] Jocher, G.; Chaurasia, A.; Qiu, J. Ultralytics YOLOv8. 2023. Available online: <https://github.com/ultralytics/ultralytics>
- [2] Liao, Z. et al. RT-DETRv4: Painlessly furthering real-time object detection with vision foundation models. 2025.
- [3] Ren, T.; Liu, S.; Zeng, A.; Lin, J.; Li, K.; Cao, H.; Chen, J.; Huang, X.; Chen, Y.; Yan, F.; et al. Grounded SAM: Assembling open-world models for diverse visual tasks. 2024.
- [4] Kirillov, A.; Mintun, E.; Ravi, N.; Mao, H.; Rolland, C.; Gustafson, L.; Xiao, T.; Whitehead, S.; Berg, A.C.; Lo, W.-Y.; et al. Segment Anything. 2023.
- [5] Damm, S.; Laszkiewicz, M.; Lederer, J.; Fischer, A. Anomaly DINO: Boosting patch-based few-shot anomaly detection with DINOv2. 2024.
- [6] Kienzle, D.; Kantarelis, K.; Barthel, K. Segformer++: Efficient token-merging strategies for high-resolution semantic segmentation. 2024.
- [7] Cavagnero, N.; Rosi, G.; Averta, G.; Lamberti, F.; Tommasi, T. PEM: Prototype-based efficient MaskFormer for image segmentation. 2024.
- [8] Hou, X.; Ma, M.; Fan, B.; Luo, P.; Wan, Y. Relation-DETR: Exploring explicit position relation prior for object detection. 2024.
- [9] Heidari, M. et al. A study on the performance of U-Net modifications in retroperitoneal tumor segmentation. 2025.

7. References

- [10] Cheng, T.; Song, L.; Ge, Y.; Liu, W.; Wang, X.; Shan, Y. YOLO-World: Real-time open-vocabulary object detection. In Proceedings of CVPR 2024; pp. 16901-16911.
- [11] Wyatt, J.; Leach, A.; Schmon, S.M.; Willcocks, C.G. AnoDDPM: Anomaly detection with denoising diffusion probabilistic models. In Proceedings of CVPR Workshops 2022; pp. 650-656.
- [12] Ren, T. et al. Grounding DINO 1.5: Advance the edge of open-set object detection. 2024.
- [13] Ravi, N.; Gabeur, V.; Hu, Y.-T.; Hu, R.; Ryali, C.; Ma, T.; Khedr, H.; Rädle, R.; Rolland, C.; Gustafson, L.; et al. SAM 2: Segment anything in images and videos. arXiv preprint, 2024.
- [14] Xiong, Y.; Varadarajan, B.; Wu, L.; Xiang, X.; Xiao, F.; Zhu, C.; Dai, X.; Wang, D.; Sun, F.; landola, F.; et al. EfficientSAM: Leveraged masked image pretraining for efficient segment anything. 2023.
- [15] Wang, C.-Y.; Yeh, I.-H.; Liao, H.-Y.M. YOLOv9: Learning what you want to learn using programmable gradient information. 2024.
- [16] Shi, R.; Li, Z.; Wu, Z.; Zhang, W.; Xu, Y.; Luo, G.; Ma, P.; Zhang, Z. An industrial carbon block instance segmentation algorithm based on improved YOLOv8. Scientific Reports, 2025.
- [17] Jiang, J.; He, Z.; Wan, A.; AL-Bukhaiti, K.; Wang, K.; Zhu, P.; Cheng, X. Zero-shot industrial anomaly detection via CLIP-DINOv2 multimodal fusion. Electronics, 2025.
- [18] Xiong, X.; Wu, Z.; Tan, S.; Li, W.; Tang, F.; Chen, Y.; Li, S.; Ma, J.; Li, G. SAM2-UNet: Segment anything 2 as a strong encoder. 2026.

7. References

- [19] Shi, C.; Wang, K.; Zhang, G.; Li, Z.; Zhu, C. Semi-supervised semantic segmentation for industrial surface defects. Scientific Reports, 2024.
- [20] Liao, K.; Nie, L.; Huang, S.; Lin, C.; Zhang, J.; Zhao, Y.; Gabbouj, M.; Tao, D. Deep learning for camera calibration and beyond: A survey. 2025.
- [21] Heckler-Kram, L.; Neudeck, J.-H.; Scheler, U.; König, R.; Steger, C. The MVTec AD 2 dataset: Advanced scenarios for anomaly detection . 2025.
- [22] Kang Liao, Lang Nie, Shujuan Huang, Chunyu Lin, Jing Zhang, Yao Zhao, Moncef Gabbouj, and Dacheng Tao. Deep learning for camera calibration and beyond: A survey, 2025.
- [23] Lars Heckler-Kram, Jan-Hendrik Neudeck, Ulla Scheler, Rebecca König, and Carsten Steger. The MVTec AD 2 dataset: Advanced scenarios for unsupervised anomaly detection, 2025.
- [24] S. Yesmin et al. Detection segmentation and noise removal process of cancerous image cell of liver through modified marker-controlled watershed approach. Springer, 2025.
- [25] A. Sen et al. Feature engineering is not dead: Reviving classical machine learning with entropy, HOG, and LBP feature fusion for image classification. arXiv preprint, 2025.
- [26] X. Xu. The application of automated image processing technology in industrial inspection. Journal of the corresponding venue, 2025.
- [27] Hua Song and Ziqiao Wen. Interactive design modeling of 3D styling combining Bezier curve and Harris corner point detection algorithm. PLOS ONE, 2025.

7. References

- [28] Ibrar Hussain. A stable region-based image segmentation model integrating fuzzy logic and geometric principles. *Acadlore Transactions on AI and Machine Learning*, 2025.
- [29] Ibrahim Almohimeed et al. Sandpiper optimization algorithm with region growing based robust retinal blood vessel segmentation approach. *IEEE Access*, 2024.
- [30] Ran Jia et al. Robust ridge-patterned instance segmentation for high-resolution point cloud measurement. *IEEE Transactions on Instrumentation and Measurement*, 2025.
- [31] Jun Guo, Yang Yang, and Yuxin Chen. Research on sub-pixel accuracy flange disk dimension measurement based on machine vision. *Signal, Image and Video Processing*, 2024.
- [32] Hewa Majeed Zangana, Ayaz Khalid Mohammed, and Firas Mahmood Mustafa. Advancements in edge detection techniques for image enhancement: A comprehensive review. *International Journal of Artificial Intelligence & Robotics*, 2024.
- [33] José L. Mira et al. Benchmarking of computer vision methods for energy-efficient high-accuracy olive fly detection on edge devices. *Multimedia Tools and Applications*, 2024.
- [34] Mikel González et al. Adaptive edge finishing process on distorted features through robot-assisted computer vision. *Journal of Manufacturing Systems*, 2024.
- [35] Hasan Muzakki et al. Evaluating K-nearest neighbors for X-ray bone fracture detection using HOG, LBP, and SIFT feature extraction methods. *IEEE Access*, 2025.
- [36] Hengxu Liu et al. Surface defect detection based on machine learning and data augmentation. *IEEE ICCASIT*, 2024.

7. References

- [37] Hafizh Eryadi. Comparative analysis of SVM and XGBoost classifiers with HOG features for concrete crack detection. IT Journal Research and Development, 2025.
- [38] Hung et al. Tool wear classification based on support vector machine and deep learning models. Sensors and Materials, 2024.
- [39] Priyanka et al. Machine learning-based phase resolved partial discharge pattern classification using HOG and PCA. IEEE Conference, 2024.
- [40] Ding et al. Homography decomposition revisited. International Journal of Computer Vision, 2025.
- [41] Zhang et al. A high-quality and convenient camera calibration method using a single image. Electronics, 2024.
- [42] Ge et al. Research on surface defect classification model of automotive flange disks based on machine vision. Signal, Image and Video Processing, 2025.
- [43] Tsung-Yi Lin et al. Microsoft COCO: Common objects in context. ECCV, 2014.
- [44] Kechen Song and Yunhui Yan. A noise robust method based on completed local binary patterns for hot-rolled steel strip surface defects. Applied Surface Science, 2013.
- [45] David R. Martin et al. A database of human segmented natural images and its application to evaluating segmentation algorithms. ICCV, 2001.

7. References

- [46] Pablo Arbeláez et al. Contour detection and hierarchical image segmentation. IEEE TPAMI, 2011.
- [47] Mark Everingham et al. The PASCAL visual object classes (VOC) challenge. IJCV, 2010.
- [48] Xiaoping Wu et al. IP102: A large-scale benchmark dataset for insect pest recognition. CVPR, 2019.
- [49] Pranav Rajpurkar et al. MURA: Large dataset for abnormality detection in musculoskeletal radiographs. arXiv, 2017.
- [50] John Canny. A computational approach to edge detection. IEEE TPAMI, 1986.
- [51] Rafael Grompone von Gioi et al. LSD: A line segment detector. Image Processing On Line, 2012.
- [52] Piotr Dollár and C. Lawrence Zitnick. Structured forests for fast edge detection. ICCV, 2013.
- [53] Luc Vincent and Pierre Soille. Watersheds in digital spaces: An efficient algorithm based on immersion simulations. IEEE TPAMI, 1991.
- [54] Alejandro F. Frangi et al. Multi-scale vessel enhancement filtering. MICCAI, 1998.